

Pet Care Tracker
A Task Management System for Pet Owners

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Project Selection

Approximately 66% of households in the United States own at least one type of pet (American Veterinary Medical Association [AVMA], 2024). Taking care of a pet is a big responsibility, and making sure all of their needs are met can be challenging. For example, tasks such as remembering whether a pet has been fed, exercised, or given their medication are easy to overlook when managing other everyday responsibilities. Wouldn't it be convenient to have one single place to organize and keep track of your pet's care? That is the purpose of this project.

Welcome to the Pet Care Tracker software system. Expanded from the concept of a common to-do list, the pet care tracker acts as a virtual agenda to manage your pet care needs. Assigning one-time or recurring tasks and creating individualized pet profiles are just the foundation of this life-changing application for pet owners. Get ready to track your pet's needs with ease using the Pet Care Tracker!

Requirements Gathering

Functional Requirements -

1. The system shall support multiple users without account creation.
2. The system shall support creation of pet profiles and tasks.
3. The system shall support assigning of tasks to specific pets.
4. The system shall support sharing of pet profiles and tasks when prompted by the user who created them.
5. The system shall support a single user managing multiple pets.
6. Upon application startup, the system shall open to the homepage.

7. The system shall list up to 3 pet profiles and today's tasks on the homepage.
8. If users create more than 3 pet profiles, the system shall populate a "See More" function to view the additional pets.
9. The system shall ensure pet profiles and tasks are saved to the local application.
10. The system shall support multiple dates and times for a single task.
11. The system shall support automatic saving of completed tasks for up to 30 days in the completed task tracker page.
12. The system shall support task and profile deletion when prompted by the user.
13. Upon completion of a task, the system shall gray and cross it out on the homepage.
14. The system shall support reversing the completion status of tasks.

Nonfunctional Requirements -

1. The system shall be available to any user in the United States.
2. The system shall be usable for users with basic computer skills.
3. The system shall support navigation to any feature with 3 clicks or less.
4. The system shall support instant task population to the homepage.
5. If the system crashes, the system shall display an error message to alert users of the error.
6. The system shall support automatic saving every 10 seconds when changes are detected.
7. The system shall support future expansion of the software system.

Architecture Design

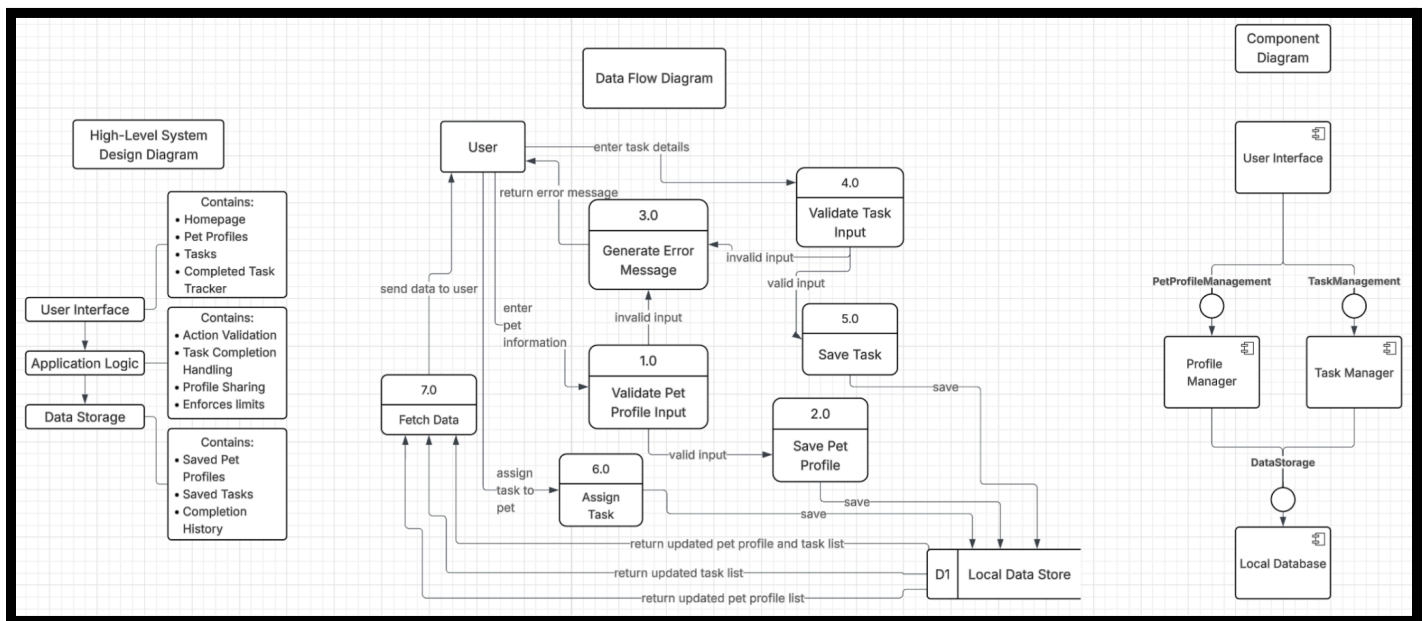
The Pet Care Tracker uses a layered, modular architecture. The system is divided into three different layers which are responsible for the user interface, application logic, and data storage. It consists of four main components, including the user interface, profile manager, task

manager, and a local database. This structure allows for scalability and adaptability for future developments of the system.

In Figure 1, the high-level design diagram illustrates the three architectural layers and demonstrates how data is exchanged between each of them. The data flow diagram shows how data flows through the system during processes such as creating pet profiles and tasks, as well as how data is updated while assigning a task to a pet. The component diagram visualizes the four main components and the interactions between them. The user interface communicates with the profile manager and task manager through provided interfaces, with both managers communicating with the local database for data storage.

Figure 1

High-Level System Architecture



Note. This figure includes the High-Level System Design (left), Data Flow Diagram (center), and Component Diagram (right).

Detailed Planning

Table 1

Project Milestones

Milestone	Description	Target Date
Milestone 1:	Planning Complete	01/20/2026
Milestone 2:	Design Complete	02/03/2026
Milestone 3:	Implementation Complete	02/14/2026
Milestone 4:	Project Complete	02/17/2026

The timeline for this project runs from January 17, 2026, until the due date on February 17, 2026. Milestone 1 consists of the completion of the planning phase of the software development life cycle. The next phase is design, with the completion of it making up milestone 2. Implementation is milestone 3 where the coding will be complete. The completion of the project marks milestone 4, ending on the due date.

The project team consists of one full stack developer who is also responsible for project management. The resources used will be Visual Studio Code as the Integrated Development Environment, and GitHub for version control. Development will take place using a MacBook Pro M4, as well as a Windows 11 desktop computer to ensure multiple operating system compatibility.

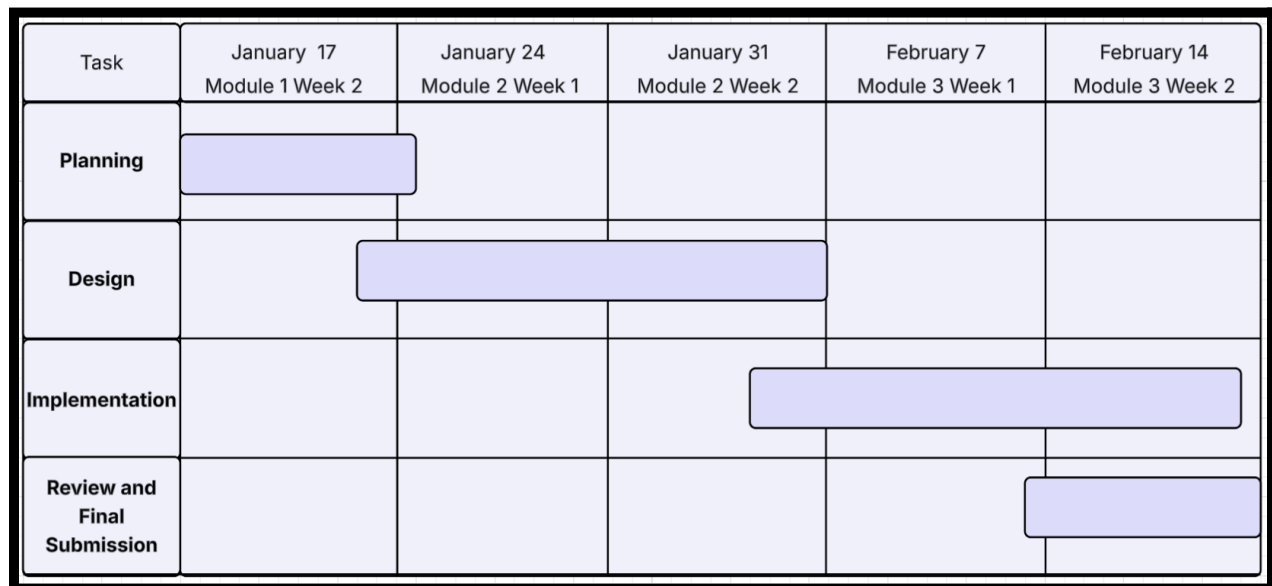
Table 2

Project Risks and Mitigations

Risk	Description	Impact	Mitigation
Time Constraints	Limited time to complete project tasks within timeline	Project delays	Adhere to the project timeline and stay within defined scope

Table 2*Project Risks and Mitigations*

Limited Technical Skills	Limited programming and software design experience	Project delays or incomplete tasks	Allocate time for tutorials and seek guidance as needed
Scope Expansion	Additional features added that were not included in the project scope.	Project delays and reduced time for other tasks	Clearly define and adhere to the original project scope
Planning Inexperience	Inexperience with project planning and design	Planning oversights leading to project delays	Thorough planning and research before implementation

Figure 2*Gantt Chart for Project Timeline*

The Gantt chart in Figure 2 illustrates the project timeline over the course of Modules 1 through 3. The chart is broken into five modules, each representing a one-week period. It details the project progression starting with Planning, and ending with the Final Submission. The

“Review and Final Submission” phase overlaps with the “Implementation” phase, ensuring that documentation is completed alongside implementation.

Proposed Implementation

The Pet Care Tracker will be developed using both Java and Python. Java will be used for the core application logic because of its object-oriented nature and maintainability. Java is also the language I am most confident with, as I am still in the process of learning Python. Python will be used to handle data-handling and testing, along with potential analytical features. The programming conventions that will be used include clear and consistent commenting, meaningful variable naming, and suitable error-handling.

Testing and Validation

Testing will be thorough to ensure the core functionality of the Pet Care Tracker is maintained. Verifying the main components are operating as intended will be the primary focus. This includes user interactions to create pet profiles and tasks, as well as assigning tasks to specific pets. Proper data storage is a critical function, so extensive testing of this aspect is crucial to ensure reliability.

Table 3

Test Cases

Test Case ID	Test Case	Objective	Steps
Test Case 1	User With Multiple Pets	Verify a single user can create more than one pet	1. Open the application 2. Verify a pet profile is listed 3. Navigate to the “Add Pet” button 4. Fill out required fields and add pet 5. Verify both pet profiles appear

Table 3*Test Cases*

Test Case 2	Homepage on Startup	Verify the application opens onto the homepage upon startup	1. Open application 2. Verify the homepage was the first view
Test Case 3	Pet Profile Creation	Verify that a user can successfully create a pet profile	1. Open the application 2. Navigate to the “Add Pet” button 3. Fill out all required fields 4. Verify the pet profile was successfully created and saved
Test Case 4	Task Creation	Verify that a user can successfully create a new task	1. Open the application 2. Navigate to the “Add Task” button 3. Fill out all required fields 4. Verify the task was successfully created and saved
Test Case 5	Task Assignment	Verify that a user can successfully assign a created task to a specific pet	1. Open the application 2. Navigate to a created pet profile 3. Navigate to the “Assign Task” button 4. Check the box for a created task 5. Verify the task appears on the selected pet’s profile
Test Case 6	Data Storage	Verify that user data is successfully stored locally	1. Open the application 2. Verify there is previous pet/task data present 3. Close the application 4. Reopen the application 5. Verify the data is still there
Test Case 7	Task Completion	Verify completed tasks	1. Open the application

Table 3*Test Cases*

History	appear on the “Completed Tasks” page	2. Checkmark a task to complete it 3. Navigate to the “Completed Tasks” page 4. Verify the completed task appears on the page
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Documentation

The Pet Care Tracker will include a user manual to guide users with system navigation. The manual will include screenshots and instructions to demonstrate the core features, such as creating a pet profile, creating and assigning tasks, completing tasks, deleting tasks or pets, and an overview of the application. Fulfilling this documentation will ensure users can properly navigate and use the system. Additionally, technical documentation will be kept up to date throughout the project with diagrams, explanations of components, descriptions for data storage functionality, and notes on future expansion. This will ensure that future developers have the tools necessary to continue work on the project.

References

American Veterinary Medical Association. (n.d.). *U.S. pet ownership statistics*. American Veterinary Medical Association. Retrieved January 20, 2026, from <https://www.avma.org/resources-tools/reports-statistics/us-pet-ownership-statistics>